

BUFFALORA, Switzerland

WMO: 067780

Lat: 46.6481N

Lon: 10.2672E

Elev: 1970

StdP: 79.79

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.5	(c) -20.8	(d) -27.0	(e) 0.4	(f) -23.1	(g) -24.0	(h) 0.5	(i) -19.5	(j) 6.8	(k) -4.2	(l) 6.0	(m) -4.4	(n) 1.1	(o) 130	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 22.8	(d) 11.9	(e) 21.1	(f) 11.5	(g) 19.4	(h) 10.9	(i) 13.3	(j) 19.0	(k) 12.5	(l) 18.4	(m) 11.8	(n) 17.5	(o) 3.1	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.7	14.2	10.5	10.0	13.3	9.6	9.5	12.6	43.9	19.5	41.4	18.3	39.4	17.8	16.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.4	(b) 4.9	(c) 4.3	(e) -27.0	(f) 25.3	(g) 2.9	(h) 1.3	(i) -29.2	(j) 26.2	(k) -30.9	(l) 27.0	(m) -32.5	(n) 27.7	(o) -34.6	(p) 28.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			(e) -27.0	(f) 25.3	(g) 2.9	(h) 1.3	(i) -29.2	(j) 26.2	(k) -30.9	(l) 27.0	(m) -32.5	(n) 27.7	(o) -34.6	(p) 28.6		
			(e) -27.6	(f) 14.8	(g) 2.9	(h) 0.9	(i) -29.7	(j) 15.5	(k) -31.4	(l) 16.0	(m) -33.1	(n) 16.5	(o) -35.2	(p) 17.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-8.3	-7.9	-3.3	1.1	4.7	9.5	11.2	11.2	7.3	3.4	-2.0	-6.8	(6)
(7)		DBStd	7.90	4.32	5.15	3.58	3.34	3.10	3.02	2.78	2.75	2.82	3.46	4.40	4.40	(7)
(8)		HDD10.0	3172	569	502	413	266	165	44	20	20	88	204	361	519	(8)
(9)		HDD18.3	6062	827	735	672	516	422	264	222	221	332	463	611	778	(9)
(10)		CDD10.0	152	0	0	0	0	2	30	56	58	6	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	27	0	0	0	0	0	12	8	7	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.7	1.6	1.6	1.8	1.8	1.9	1.8	1.8	1.7	1.6	1.6	1.6	1.3	(14)
(15)	Precipitation	PrecAvg	749	33	31	35	46	64	88	97	98	69	79	72	37	(15)
(16)		PrecMax	1024	84	111	94	92	139	164	190	164	188	239	280	106	(16)
(17)		PrecMin	563	6	7	4	9	15	33	53	31	13	4	7	2	(17)
(18)		PrecStd	125	22	23	21	22	29	35	29	32	45	51	64	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	8.2	10.6	13.7	19.0	25.1	24.5	24.4	19.8	17.3	12.1	4.7	(19)
(20)			MCWB	0.5	1.3	2.2	5.3	8.6	11.9	12.2	12.2	9.8	7.5	3.7	0.4	(20)
(21)		2%	DB	2.4	5.1	8.4	12.2	16.3	22.2	22.4	22.6	18.4	14.2	8.2	2.8	(21)
(22)			MCWB	-0.8	-0.1	1.1	4.4	7.8	11.2	11.7	12.4	9.8	6.6	2.5	-0.4	(22)
(23)		5%	DB	0.4	2.3	6.5	10.4	14.3	20.0	20.7	20.7	16.6	12.0	5.5	1.2	(23)
(24)			MCWB	-1.5	-1.3	0.5	3.8	7.1	10.8	11.1	12.1	9.3	5.8	2.0	-0.6	(24)
(25)		10%	DB	-1.1	-0.1	4.5	8.3	12.2	17.7	18.9	18.9	14.6	9.9	3.7	-0.1	(25)
(26)			MCWB	-2.5	-2.2	-0.3	3.0	6.0	9.8	10.6	11.2	8.4	5.7	1.7	-1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.2	3.1	6.1	9.9	13.0	14.6	14.6	11.9	8.8	5.3	1.8	(27)
(28)			MCDWB	3.4	5.9	8.1	10.8	15.8	19.8	19.5	20.7	16.6	13.4	9.0	3.2	(28)
(29)		2%	WB	0.0	0.5	2.0	5.3	8.8	12.1	13.3	13.4	10.8	7.7	3.9	0.7	(29)
(30)			MCDWB	1.7	3.7	7.1	10.4	14.2	20.1	18.9	19.2	16.1	11.5	6.6	1.7	(30)
(31)		5%	WB	-1.2	-0.9	1.0	4.3	7.6	11.4	12.5	12.6	10.0	7.0	2.8	-0.3	(31)
(32)			MCDWB	0.1	1.8	5.1	8.7	12.3	18.8	18.1	18.2	14.7	10.0	4.6	0.8	(32)
(33)		10%	WB	-2.5	-2.1	0.1	3.2	6.6	10.5	11.6	11.9	9.2	6.3	1.8	-1.5	(33)
(34)			MCDWB	-1.2	-0.2	3.7	6.9	10.7	16.7	17.0	17.2	13.3	9.0	3.5	-0.1	(34)
(35)	Mean Daily Temperature Range	MDBR		12.2	13.7	13.2	11.3	10.9	12.7	13.3	13.1	12.4	11.7	10.3	11.9	(35)
(36)		5% DB	MCDBR	13.0	17.1	16.9	15.3	15.7	17.6	17.0	16.4	16.1	15.9	13.2	12.0	(36)
(37)			MCWBR	10.3	12.0	10.3	8.3	7.6	7.5	7.2	7.3	8.2	8.9	8.2	9.0	(37)
(38)			MCDBR	11.2	13.6	13.2	12.5	13.8	16.0	14.4	14.3	13.8	12.4	9.8	9.8	(38)
(39)		5% WB	MCWBR	9.3	9.8	8.2	7.2	7.1	7.1	6.8	6.9	7.5	7.7	6.5	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.217	0.238	0.274	0.316	0.330	0.337	0.335	0.329	0.302	0.279	0.246	0.217	(40)
(41)		taud		2.371	2.320	2.294	2.313	2.380	2.399	2.403	2.439	2.503	2.544	2.509	2.408	(41)
(42)		Ebn at Noon		916	955	957	940	934	925	923	917	918	898	874	877	(42)
(43)		Edh at Noon		68	90	109	119	116	115	113	104	89	73	61	59	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.59	2.54	3.76	4.49	4.88	5.60	5.63	4.91	3.80	2.54	1.55	1.31	(44)
(45)		RadStd		0.18	0.29	0.47	0.64	0.52	0.45	0.44	0.45	0.38	0.32	0.24	0.17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (104 neighbors)		+0.34	N/A	N/A	+0.64	+0.42	+0.36	N/A	N/A	+59	+16		

Nomenclature: See separate page